Business Case Study – Data Warehouse Analytics

1. Changes Over Time Analysis

Query Purpose

This analysis examines how the business performance evolves over time by calculating yearly sales revenue, total customers, and total quantity sold. It helps identify growth trends and overall business expansion.

SQL Query

```
SELECT
    DATETRUNC(YEAR, order_date) AS order_year,
    SUM(sales_amount) AS total_sales,
    COUNT(DISTINCT customer_key) AS total_customers,
    SUM(quantity) AS total_quantity
FROM gold.fact_sales
WHERE order_date IS NOT NULL
GROUP BY DATETRUNC(YEAR, order_date)
ORDER BY DATETRUNC(YEAR, order_date);
```

Output Summary

Year	Total Sales	Customers	Quantity
2010	43,419	14	14
2011	707,508	2,216	2,216
2012	584,231	3,255	3,397
2013	1,634,878	17,427	52,807
2014	45,642	834	1,970

Insights

- **Rapid business growth from 2010 → 2013**
  Sales increased dramatically from **43K to 1.63M**, indicating strong business expansion.
- **Customer base growth**
  Customers increased from **14 to 17K**, suggesting successful acquisition strategies.
- **High product demand in 2013**
  Quantity sold jumped significantly to **52K units**, indicating peak demand.
- **Drop in 2014**
  Sales dropped sharply, likely because the dataset contains **partial year data**.

Action Items

✓ Increase marketing investments during high-growth periods.

✓ Identify which **products drove the 2013 sales spike** and replicate similar strategies.

✓ Investigate reasons for lower sales in later periods:

- product availability
- marketing campaigns
- supply chain issues.

✓ Forecast future demand based on historical growth trends.

2. Cumulative Sales Analysis

Query Purpose

This analysis calculates cumulative sales performance over time and tracks price trends using running totals and moving averages.

This helps answer:

- How total revenue accumulates over time
- Whether product prices are increasing or decreasing

SQL Query

```
SELECT
order_date,
total_sales,
SUM(total_sales) OVER (ORDER BY order_date) AS running_total_sales,
AVG(avg_price) OVER (ORDER BY order_date) AS moving_average_price
FROM
(
```

```
SELECT
DATETRUNC(YEAR, order_date) AS order_date,
SUM(sales_amount) AS total_sales,
AVG(price) AS avg_price
FROM gold.fact_sales
WHERE order_date IS NOT NULL
GROUP BY DATETRUNC(YEAR, order_date)
) AS subquery;
```

Output Summary

Year	Total Sales	Running Total Sales	Moving Avg Price
2010	43,419	43,419	3101
2011	707,508	711,507	3146
2012	584,231	1,296,738	2670
2013	1,634,878	2,930,516	2080
2014	45,642	2,935,158	1668

Insights

- **Strong cumulative revenue growth**
  Total revenue steadily increases year by year, reaching nearly **2.9M cumulative sales by 2014**.
- **Peak contribution in 2013**
  The majority of the revenue growth occurs in **2013**, indicating a significant business expansion.
- Declining price trend
  The moving average price drops from ~3100 → ~1668, suggesting:
  - price reductions
  - product mix changes
  - promotions or discounts.
- **Stable revenue accumulation**
  Despite price decreases, sales volume continued to grow until 2013.

Action Items

✓ Analyze product categories responsible for high revenue growth.

✓ If prices are declining:

- evaluate pricing strategy
- ensure margins remain profitable.

✓ Investigate if discounts or promotions caused the lower average price.

✓ Focus on high-volume products that drive cumulative sales growth.

3. Product Sales Performance Analysis

Query Purpose

This analysis evaluates product-level sales performance over time by comparing yearly sales with historical averages and previous-year sales.

This helps answer:

Which products are performing above or below their historical average

Which products show year-over-year (YoY) growth or decline

Which products are gaining or losing market demand

SQL Query:

```
WITH yearly_product_sales AS (
    SELECT
        YEAR(f.order_date) AS order_year,
        p.product_name,
        SUM(f.sales_amount) AS current_sales
    FROM gold.fact_sales f
    LEFT JOIN gold.dim_products p
        ON f.product_key = p.product_key
    WHERE f.order_date IS NOT NULL
    GROUP BY
        YEAR(f.order_date),
        p.product_name
)

SELECT
    order_year,
    product_name,
    current_sales,
    AVG(current_sales) OVER (PARTITION BY product_name) AS avg_sales,
    current_sales - AVG(current_sales) OVER (PARTITION BY product_name) AS
diff_avg,
    CASE
        WHEN current_sales - AVG(current_sales) OVER (PARTITION BY product_name) >
0 THEN 'Above Avg'
        WHEN current_sales - AVG(current_sales) OVER (PARTITION BY product_name) <
0 THEN 'Below Avg'
        ELSE 'Avg'
    END AS avg_change,

-- YoY Analysis
    COALESCE(LAG(current_sales) OVER (PARTITION BY product_name ORDER BY
order_year),0) AS py_sales,

    COALESCE(current_sales - LAG(current_sales)
    OVER (PARTITION BY product_name ORDER BY order_year),0) AS diff_py,

    CASE
        WHEN current_sales - LAG(current_sales) OVER (PARTITION BY product_name
ORDER BY order_year) > 0 THEN 'Increase'
```

```
        WHEN current_sales - LAG(current_sales) OVER (PARTITION BY product_name
ORDER BY order_year) < 0 THEN 'Decrease'
        ELSE 'No Change'
    END AS py_change

FROM yearly_product_sales
ORDER BY product_name, order_year;
```

Insights

1. Identification of High Performing Products

Products labeled as Above Avg indicate years where sales exceeded their historical average, highlighting periods of strong demand.

2. Detection of Sales Growth Trend

The YoY comparison identifies products with increasing sales momentum. Products marked as Increase demonstrate positive growth compared to the previous year.

3. Early Detection of Declining Product Demand

Products with Below Avg performance and YoY Decrease may indicate declining demand, potential market saturation, or increased competition.

4. Sales Variability Across Years

Certain products show significant fluctuations in sales across years, suggesting possible influences such as marketing campaigns, promotions, or seasonal demand patterns.

Action Items

✓ Prioritize marketing efforts for products showing **consistent YoY growth**.

✓ Investigate reasons behind products experiencing **sales decline**.

✓ Replicate successful strategies used during **high-performing years**.

✓ Optimize inventory planning for products showing **consistent above-average demand**.

4. Part-To-Whole Analysis

Query Purpose

This analysis evaluates how each **product category contributes to overall business performance** by measuring the share of total sales and the distribution of customers across

categories. It helps identify which categories drive revenue and which categories attract the most customers, enabling better product strategy and cross-selling opportunities.

SQL Query

```
--4. Part-To-Whole Analysis
---4.1 Based on Sales
WITH category_sales AS (
    SELECT
        p.category,
        SUM(f.sales_amount) AS total_sales
    FROM gold.fact_sales f
    LEFT JOIN gold.dim_products p
        ON p.product_key = f.product_key
    GROUP BY p.category
)

SELECT
    category,
    total_sales,
    SUM(total_sales) OVER () AS overall_sales,
    CONCAT(
        ROUND((CAST(total_sales AS FLOAT) / SUM(total_sales) OVER ()) *
100, 2),
        '%'
    ) AS percentage_of_total
FROM category_sales
ORDER BY total_sales DESC;

---4.2 Based on Customers
WITH category_sales AS (
    SELECT
        p.category,
        COUNT(DISTINCT f.customer_key) AS total_customers
    FROM gold.fact_sales f
    LEFT JOIN gold.dim_products p
        ON p.product_key = f.product_key
    GROUP BY p.category
)

SELECT
    category,
    total_customers,
    SUM(total_customers) OVER () AS overall_customers,
    CONCAT(
        ROUND((CAST(total_customers AS FLOAT) / SUM(total_customers) OVER
()) * 100, 2),
        '%'
    ) AS percentage_of_total
FROM category_sales
ORDER BY total_customers DESC;
```

Insights

➤ **Revenue concentration in the Bikes category**
The Bikes category contributes **96.46% of total sales**, indicating that the business revenue is heavily dependent on bike products. This suggests that bikes are the primary revenue driver of the company.

➤ **Accessories and Clothing contribute minimal revenue**
Accessories account for **2.39% of total sales**, while Clothing contributes **1.16%**, indicating these categories currently play a supporting role rather than being major revenue contributors.

➤ **Accessories attract the highest number of customers**
Accessories account for **48.6% of total customers**, which suggests that these products are frequently purchased and may serve as entry-level or complementary purchases.

➤ **Bikes generate higher revenue per customer**
Although Bikes represent **29.37% of total customers**, they contribute the majority of revenue. This indicates that bike purchases involve **higher transaction values compared to other categories**.

➤ **Clothing represents a moderate customer segment**
Clothing accounts for **22.03% of customers**, indicating moderate engagement but relatively lower revenue contribution compared to bikes.

Action Items

✓ **Strengthen cross-selling strategies**
Promote accessories and clothing alongside bike purchases to increase overall order value.

✓ **Bundle products with bikes**
Create bundled offers such as bike + helmet + accessories to increase accessory and clothing sales.

✓ **Leverage accessories as entry products**
Since accessories attract the largest customer share, they can be used as gateway products to introduce customers to higher-value products like bikes.

✓ **Improve visibility of clothing products**
Increase promotions, marketing visibility, and product placement to improve clothing sales contribution.

✓ **Reduce dependency on a single category**
Since revenue is highly concentrated in bikes, expanding accessory and clothing product lines can help diversify revenue sources.

5. Data Segmentation

Query Purpose

This analysis segments products and customers into meaningful groups to better understand product pricing distribution and customer purchasing behavior. Product segmentation based on cost helps identify how products are distributed across different price ranges, while customer segmentation based on spending behavior helps categorize customers according to their purchasing value and engagement over time.

SQL Query

```
--5. Data Segmentation
---5.1 Based on Cost Range
WITH product_segments AS (
    SELECT
        product_key,
        product_name,
        cost,
        CASE
            WHEN cost < 100 THEN 'Below 100'
            WHEN cost BETWEEN 100 AND 500 THEN '100-500'
            WHEN cost BETWEEN 500 AND 1000 THEN '500-1000'
            ELSE 'Above 1000'
        END AS cost_range
    FROM gold.dim_products
)

SELECT
    cost_range,
    COUNT(product_key) AS total_products
FROM product_segments
GROUP BY cost_range
ORDER BY total_products DESC;

---5.2 Based on Customer Spending behaviour
WITH customer_spending AS (
    SELECT
        c.customer_key,
        SUM(f.sales_amount) AS total_spending,
        MIN(f.order_date) AS first_order,
        MAX(f.order_date) AS last_order,
        DATEDIFF(month, MIN(f.order_date), MAX(f.order_date)) AS lifespan
    FROM gold.fact_sales f
    LEFT JOIN gold.dim_customers c
        ON f.customer_key = c.customer_key
    GROUP BY c.customer_key
)

SELECT
    customer_segment,
    COUNT(customer_key) AS total_customers
FROM (
    SELECT
        customer_key,
        CASE
            WHEN lifespan >= 12 AND total_spending > 5000 THEN 'VIP'
```

```
            WHEN lifespan >= 12 AND total_spending <= 5000 THEN 'Regular'
            ELSE 'New'
        END AS customer_segment
    FROM customer_spending
) t
GROUP BY customer_segment
ORDER BY total_customers DESC;
```

Insights

➤ **Majority of products fall in the lower price ranges**

The Below 100 cost range contains 110 products, making it the largest segment. This indicates that the product portfolio is heavily focused on affordable or entry-level products.

➤ **Mid-range products also represent a significant portion of the catalog**

The 100-500 range includes 101 products, showing that the business also maintains a strong presence in the mid-price segment.

➤ **Premium products are relatively limited**

Only 45 products fall in the 500-1000 range and 39 products are priced above 1000, suggesting that premium offerings represent a smaller portion of the product portfolio.

➤ **Majority of customers are new customers**

The segmentation shows 14,631 customers classified as New, indicating that most customers have either recently started purchasing or have limited purchase history.

➤ **Smaller but valuable loyal customer segments exist**

There are 2,198 Regular customers and 1,655 VIP customers, representing loyal and higher-value customers who contribute significantly to long-term revenue.

Action Items

✓ **Strengthen marketing for mid-range and premium products**
Since most products are low-priced, promoting mid-range and premium products may help increase overall revenue and profit margins.

✓ **Use entry-level products to acquire new customers**
Low-cost products can be used as entry points to attract new customers and introduce them to the brand.

✓ **Implement targeted loyalty programs for VIP customers**
Provide exclusive benefits such as discounts, early product access, or loyalty rewards to retain high-value customers.

✓ **Develop retention strategies for new customers**
With a large number of new customers, strategies such as personalized offers, onboarding promotions, and follow-up campaigns can help convert them into regular buyers.

✓ **Encourage product upgrades and cross-selling**
Promote mid-range or premium products to existing customers who initially purchase lower-priced items to increase average order value.

6. Customer Reporting View

Query Purpose

This section builds a **customer reporting view** that consolidates customer demographic information, purchasing behavior, and key performance metrics into a single analytical dataset. The view simplifies customer analysis by combining transactional data with customer attributes and computing metrics such as total orders, total sales, customer lifespan, recency, average order value, and average monthly spending. This structured view can be directly used for reporting, dashboards, and advanced customer analytics.

SQL Query

```
--6. Build Customer Reports
GO
CREATE VIEW gold.report_customers AS
WITH base_query AS (
/* ---------------------------------------------------------------------------
   1) Base Query: Retrieves core columns from tables
--------------------------------------------------------------------------- */
    SELECT
        f.order_number,
        f.product_key,
        f.order_date,
        f.sales_amount,
        f.quantity,
        c.customer_key,
        c.customer_number,
        CONCAT(c.first_name, ' ', c.last_name) AS customer_name,
        DATEDIFF(year, c.birthdate, GETDATE()) AS age
    FROM gold.fact_sales f
    LEFT JOIN gold.dim_customers c
        ON c.customer_key = f.customer_key
    WHERE f.order_date IS NOT NULL
),

customer_aggregation AS(
/* ---------------------------------------------------------------------------
2) Customer Aggregations: Summarizes key metrics at the customer level
--------------------------------------------------------------------------- */
SELECT
    customer_key,
    customer_number,
    customer_name,
    age,
    COUNT(DISTINCT order_number) AS total_orders,
```

```
    SUM(sales_amount) AS total_sales,
    SUM(quantity) AS total_quantity,
    COUNT(DISTINCT product_key) AS total_products,
    MAX(order_date) AS last_order_date,
    DATEDIFF(month, MIN(order_date), MAX(order_date)) AS lifespan
FROM base_query
GROUP BY
    customer_key,
    customer_number,
    customer_name,
    age)

SELECT
    customer_key,
    customer_number,
    customer_name,
    age,
    CASE
        WHEN age < 20 THEN 'Under 20'
        WHEN age BETWEEN 20 AND 29 THEN '20-29'
        WHEN age BETWEEN 30 AND 39 THEN '30-39'
        WHEN age BETWEEN 40 AND 49 THEN '40-49'
        ELSE '50 and above'
    END AS age_group,
    CASE
        WHEN lifespan >= 12 AND total_sales > 5000 THEN 'VIP'
        WHEN lifespan >= 12 AND total_sales <= 5000 THEN 'Regular'
        ELSE 'New'
    END AS customer_segment,
    last_order_date,
    total_orders,
    total_sales,
    total_quantity,
    total_products,
    DATEDIFF(month,last_order_date,GETDATE()) AS recency,
    CASE WHEN total_sales = 0 THEN 0
         ELSE total_sales/total_orders
    END AS avg_order_value,
    CASE WHEN lifespan = 0 THEN total_sales
         ELSE total_sales/lifespan
    END AS avg_monthly_spend
FROM customer_aggregation;
```

Insights

➤ Consolidated customer analytics dataset

The view integrates **customer demographic data and transaction data** into a single dataset, making it easier to analyze customer behavior without repeatedly joining multiple tables.

➤ Customer lifecycle and engagement tracking

Metrics such as **lifespan and recency** allow businesses to track how long customers stay active and how recently they made a purchase, which helps identify active and inactive customers.

➤ Customer value segmentation

Customers are categorized into **VIP, Regular, and New segments** based on their purchase history and spending patterns, enabling targeted marketing strategies.

➤ Purchase behavior metrics for each customer
Metrics such as **total orders, total sales, and total products purchased** help evaluate customer purchasing patterns and overall contribution to business revenue.

➤ Customer spending efficiency indicators
Calculated metrics like **average order value and average monthly spend** help determine how much revenue each customer generates per order and over time.

Action Items

✓ Use the reporting view as a centralized dataset for dashboards
The view can serve as a **semantic layer for BI tools such as Power BI or Tableau**, reducing the need for complex joins during reporting.

✓ Implement targeted marketing campaigns based on customer segments
VIP customers can receive exclusive offers and loyalty rewards, while new customers can be targeted with onboarding promotions.

✓ Identify high-value customers and retention opportunities
Customers with high spending and long lifespans should be prioritized for retention strategies such as loyalty programs.

✓ Detect inactive or declining customers using recency metrics
Customers with high recency values may require re-engagement campaigns such as personalized offers or reminders.

✓ Monitor customer purchase trends over time
Metrics like average order value and monthly spending help track changes in customer purchasing behavior and adjust pricing or promotion strategies accordingly.

7. Product Performance Analysis

Query Purpose

This analysis evaluates product performance by combining product information with sales transactions and calculating key performance indicators at the product level. The objective is to understand how individual products contribute to overall sales, customer engagement, and revenue generation. It also identifies high-performing and low-performing products using sales-based segmentation.

SQL Query

--1.Base Product Information

```
SELECT
    p.product_key,
    p.product_name,
    p.category,
    p.subcategory,
    p.cost
FROM gold.dim_products p


--2.Join with Sales Data
SELECT
    p.product_key,
    p.product_name,
    p.category,
    p.subcategory,
    p.cost
FROM gold.dim_products p
LEFT JOIN gold.fact_sales s
ON p.product_key = s.product_key;


--3. Aggregate Sales Metrics

SELECT
    p.product_key,
    p.product_name,
    p.category,
    p.subcategory,
    p.cost,

    COALESCE(COUNT(DISTINCT s.order_number),0) AS total_orders,

    COALESCE(SUM(s.sales_amount),0) AS total_sales,

    COALESCE(SUM(s.quantity),0) AS total_quantity_sold,

    COALESCE(COUNT(DISTINCT s.customer_key),0) AS total_customers,

    COALESCE(DATEDIFF(month, MIN(s.order_date), MAX(s.order_date)),0) AS
lifespan_months,

    COALESCE(DATEDIFF(month, MAX(s.order_date), GETDATE()),0) AS recency_months,

    COALESCE(SUM(s.sales_amount) * 1.0 / COUNT(DISTINCT s.order_number),0) AS
avg_order_revenue,

    COALESCE(SUM(s.sales_amount) * 1.0 /
    NULLIF(DATEDIFF(month, MIN(s.order_date), MAX(s.order_date)),0),0)
    AS avg_monthly_revenue,

    CASE
    WHEN SUM(s.sales_amount) > 50000 THEN 'High Performer'
    WHEN SUM(s.sales_amount) > 10000 THEN 'Mid Range'
    ELSE 'Low Performer'
```

```
    END AS product_segment

FROM gold.dim_products p
LEFT JOIN gold.fact_sales s
    ON p.product_key = s.product_key

GROUP BY
    p.product_key,
    p.product_name,
    p.category,
    p.subcategory,
    p.cost

ORDER BY total_sales DESC;
```

Insights

➤ **Centralized product performance view**
The query consolidates product attributes and sales transactions to produce a comprehensive product performance dataset that includes revenue, customer engagement, and sales volume metrics.

➤ **Identification of top-performing products**
Products categorized as **High Performer** generate significant revenue and contribute strongly to overall business performance.

➤ **Detection of low-performing products**
Products categorized as **Low Performer** may indicate limited demand, poor visibility, or outdated product offerings.

➤ **Product lifecycle visibility**
Metrics such as **lifespan months and recency months** help identify products that have been consistently sold over time versus products that may have declining demand.

➤ **Revenue efficiency measurement**
Metrics such as **average order revenue** and **average monthly revenue** help determine how efficiently each product contributes to revenue generation.

Action Items

✓ **Promote high-performing products**
Products classified as **High Performer** should be prioritized in marketing campaigns, featured placements, and promotional strategies.

✓ **Improve performance of mid-range products**
Products categorized as **Mid Range** may benefit from targeted promotions, pricing adjustments, or improved visibility.

✓ **Review underperforming products**
Products classified as **Low Performer** should be analyzed to determine whether they require pricing changes, marketing support, or product improvements.

✓ **Optimize inventory planning**
Products with higher sales frequency and strong revenue metrics should receive priority in inventory management and supply planning.

✓ **Use product insights for cross-selling strategies**
Products frequently purchased together can be bundled or recommended to customers to increase overall order value.

8. Product Reporting View

Query Purpose

This section creates a **product reporting view** that consolidates product information and sales transactions into a single analytical dataset. The view calculates key product performance metrics such as total sales, total orders, customer reach, product lifespan, recency, and revenue efficiency. It enables analysts and business users to easily evaluate product performance and identify high-performing and underperforming products for strategic decision-making.

SQL Query

```
GO
CREATE VIEW gold.report_products AS

WITH base_query AS (
    SELECT
        f.order_number,
        f.order_date,
        f.customer_key,
        f.sales_amount,
        f.quantity,
        p.product_key,
        p.product_number,
        p.product_name,
        p.category,
        p.subcategory,
        p.cost
    FROM gold.fact_sales f
    LEFT JOIN gold.dim_products p
        ON p.product_key = f.product_key
    WHERE f.order_date IS NOT NULL
),

product_aggregation AS (

SELECT
    product_key,
```

```
    product_number,
    product_name,
    category,
    subcategory,
    cost,

    COUNT(DISTINCT order_number) AS total_orders,
    SUM(sales_amount) AS total_sales,
    SUM(quantity) AS total_quantity,
    COUNT(DISTINCT customer_key) AS total_customers,

    MIN(order_date) AS first_sale_date,
    MAX(order_date) AS last_sale_date,

    DATEDIFF(month, MIN(order_date), MAX(order_date)) AS lifespan

FROM base_query

GROUP BY
    product_key,
    product_number,
    product_name,
    category,
    subcategory,
    cost
)

SELECT
    product_key,
    product_number,
    product_name,
    category,
    subcategory,
    cost,

    total_orders,
    total_sales,
    total_quantity,
    total_customers,

    first_sale_date,
    last_sale_date,

    lifespan,

    DATEDIFF(month, last_sale_date, GETDATE()) AS recency,

    CASE
        WHEN total_orders = 0 THEN 0
        ELSE total_sales / total_orders
    END AS avg_order_revenue,

    CASE
        WHEN lifespan = 0 THEN total_sales
        ELSE total_sales / lifespan
    END AS avg_monthly_revenue,

    CASE
        WHEN total_sales > 50000 THEN 'High Performer'
        WHEN total_sales > 10000 THEN 'Mid Range'
        ELSE 'Low Performer'
    END AS product_segment

FROM product_aggregation;
```

Insights

➤ **Consolidated product performance dataset**
The view centralizes product attributes and sales metrics, making it easier to analyze product performance without repeatedly joining multiple tables.

➤ **Product lifecycle visibility**
Metrics such as **first sale date, last sale date, lifespan, and recency** help monitor product lifecycle stages and identify declining or inactive products.

➤ **Product revenue efficiency tracking**
Metrics like **average order revenue and average monthly revenue** allow businesses to evaluate how efficiently each product generates revenue.

➤ **Product performance segmentation**
Products are classified into **High Performer, Mid Range, and Low Performer**, enabling quick identification of top-selling and underperforming products.

Action Items

✓ **Use the view as a product analytics dataset for dashboards**
The view can serve as a **direct source for BI tools** such as Power BI or Tableau.

✓ **Promote high-performing products**
Products categorized as **High Performer** should receive greater marketing visibility and promotional support.

✓ **Monitor declining products**
Products with high recency values or low sales may require marketing support or product improvement strategies.

✓ **Optimize product portfolio decisions**
Insights from this view can help guide pricing strategies, product promotions, and inventory planning.